

Practical 3 - Parametric regression modelling

Survival Analysis

London School of Hygiene & Tropical Medicine

SOLUTIONS

Datasets required: `whitehall`

R packages required: `survival`, `mexhaz`, `flexsurv`

```
library(survival)
library(mexhaz)
library(flexsurv)
```

Introduction

In this practical we will use the Whitehall data, which is familiar from the earlier practicals. We will investigate the association between job grade (`grade`) and risk of coronary heart disease (`chd`), with and without adjustment for age. We will use the time-in-study timescale except for in one question.

Variable	Description
<code>id</code>	Unique identifier for each participant
<code>timebth</code>	Date of birth
<code>timein</code>	Date person entered the study
<code>timeout</code>	Date of the end of follow-up due to either death or censoring
<code>chd</code>	Event indicator at the end of follow-up: 0=alive or died from other cause, 1=death due to coronary heart disease
<code>grade</code>	Job grade at study entry. 1=professional/executive; 2=clerical/other
<code>agein</code>	Age in years at study entry

Load the data using:

```
whl<-read.table("datasets/whitehall.csv",sep=" ",header=T)
```

Aims

The aims of this session are:

- Fit exponential and Weibull distribution models to survival data and interpret the results.
- Use plots to assess whether a Weibull model is appropriate.
- Assess the constant hazard assumption of the exponential model.
- Understand the effect of changing the analysis timescale on estimates from the exponential model.

Question 1

We will begin with non-parametric analyses.

- Format the dates as in Practicals 1 and 2.
- How many individuals are there in each job grade group?
- Use a Kaplan-Meier plot to compare survival in the two groups. Interpret the plots. What is the estimated probability of survival beyond 10 years in each job grade group?
- Use the log rank test to compare the estimated survivor curves in the two job grades.

SOLUTION

```
#formatting dates
whl$timein=as.Date(whl$timein,"%d%b%Y")
whl$timeout=as.Date(whl$timeout,"%d%b%Y")
whl$timebth=as.Date(whl$timebth,"%d%b%Y")

#summarising job grade
table(whl$grade)
```

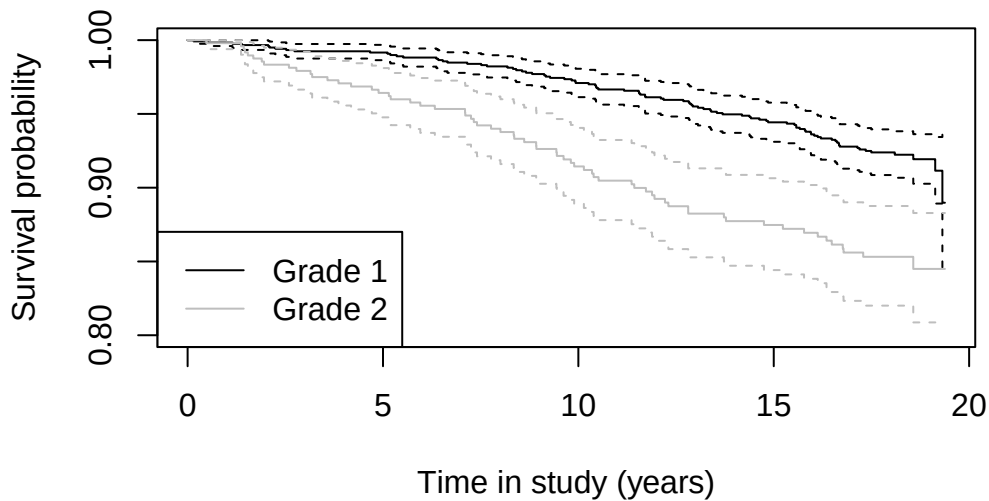
```
      1      2
1194  483
```

```
prop.table(table(whl$grade))
```

1 2
0.7119857 0.2880143

```
#Kaplan-Meier analysis
whl.km <- survfit(Surv(time=(timeout-timein)/365.25,event=chd)~grade,
                  data=whl)

plot(whl.km,ylim=c(0.8,1),conf.int = T,
     col=c("black","grey"),
     xlab="Time in study (years)", ylab="Survival probability")
legend (x="bottomleft",
       c("Grade 1","Grade 2"),
       col=c("black","grey"),lty=1)
```



```
#Estimates at 10 years
summary(whl.km,times=10)
```

Call: survfit(formula = Surv(time = (timeout - timein)/365.25, event = chd) ~ grade, data = whl)

	time	grade=1 n.risk	n.event	survival	std.err	lower 95% CI
	1.00e+01	1.11e+03	3.40e+01	9.71e-01	4.91e-03	9.61e-01
upper 95% CI						9.81e-01

	time	grade=2 n.risk	n.event	survival	std.err	lower 95% CI
	10.0000	383.0000	39.0000	0.9144	0.0131	0.8890
upper 95% CI	0.9405					

```
#log rank test
survdif(Surv(time=(timeout-timein)/365.25,event=chd)~grade,data=whl)
```

Call:

```
survdif(formula = Surv(time = (timeout - timein)/365.25, event = chd) ~
        grade, data = whl)
```

	N	Observed	Expected	(O-E) ² /E	(O-E) ² /V
grade=1	1194	90	114.2	5.12	19.9
grade=2	483	64	39.8	14.70	19.9

Chisq= 19.9 on 1 degrees of freedom, p= 8e-06

- Of the 1677 participants, 483 were in job grade 1 (29%) and 1194 were in job grade 2 (71%) at the time of entering the study.
- Those in grade 1 had better survival compared with those in grade 2. The confidence intervals are non-overlapping, suggesting evidence of a survival difference between the two groups.
- The estimated probability of survival beyond 10 years is 0.97 for grade 1 (95% CI 0.96 to 0.98) and 0.91 for grade 2 (95% CI 0.89 to 0.94).
- The p-value from the log rank test is very small. There is strong evidence against the null hypothesis that the survival curves are the same in the two job grades.

Question 2

We will now fit an exponential model to the Whitehall data using job grade as an explanatory variable.

- Write down the hazard model.
- Fit the exponential model using `mexhaz` using the code shown below.
- Identify and interpret the hazard ratio estimate.
- Identify and interpret the estimate of λ .

```
#Fitting exponential model using mexhaz
whl$time<-as.numeric(whl$timeout-whl$timein)/365.25
whl.exp<-mexhaz(Surv(time=time,event=chd)~as.factor(grade),
               base="pw.cst",data=whl,print.level = 0)
summary(whl.exp)
```

SOLUTION

- The form of the hazard model is $h(t|X) = \lambda e^{\beta X}$ where $X = 0$ for job grade 1 and $X = 1$ for job grade 2.

```
#Fitting exponential model using mexhaz
whl$time<-as.numeric(whl$timeout-whl$timein)/365.25
whl.exp<-mexhaz(Surv(time=time,event=chd)~as.factor(grade),
               base="pw.cst",data=whl,print.level = 0)
```

Data

Name	N.Obs.Tot	N.Obs	N.Events	N.Clust
whl	1677	1677	154	1

Details

Iter	Eval	Base	Nb.Leg	Nb.Aghq	Optim	Method	Code	LogLik	Total.Time
7	10	pw.cst	20	10	nlm	---	1	-944.6851	0.01

```
summary(whl.exp)
```

Call:

```
mexhaz(formula = Surv(time = time, event = chd) ~ as.factor(grade),
       data = whl, base = "pw.cst", print.level = 0)
```

Coefficients:

	Estimate	StdErr	z.value	p.value
[0,19.4]	-5.42045	0.10541	-51.4229	< 2.2e-16 ***
as.factor(grade)2	0.68850	0.16351	4.2107	2.546e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Note: For a piecewise constant baseline hazard, the coefficients correspond to the logarithm of the hazard on each time interval.

Hazard ratios (for proportional effect variables):

	Coef	HR	CI.lower	CI.upper
as.factor(grade)2	0.6885	1.9907	1.4449	2.7428

log-likelihood: -944.6851 (for 2 degree(s) of freedom)

number of observations: 1677, number of events: 154

- The estimated hazard ratio (e^β) is 1.99, meaning that the hazard rate in grade 2 was nearly two times that in grade 1. The 95% confidence interval for HR is (1.44, 2.74), indicating strong evidence against the null hypothesis that job grade is not associated with the hazard.
- The constant term is the estimate of the hazard for individuals in job grade 1, i.e. the estimate of λ . The estimate from the model is $\hat{\lambda} = e^{-5.42} = 0.0044$. The interpretation is that we expect 0.0044 events per one person year of follow-up, or 4.4 events per 100 person years of follow-up, for individuals in job grade 1.

Question 3

- Using the estimates of λ and β in model output above, calculate (by hand) an estimate of the probability of survival beyond 10 years in the two job grade groups.
- Check your answer using the code below.
- If you have extra time at the end, have a look at the examples in the lecture notes to see other ways in which exponential models can be fitted in R, and how corresponding survival curves can be obtained.

```
#Obtain survival probabilities
surv.grade1.exp<-predict(whl.exp,
                          data.val = data.frame(grade=1),
                          time.pts = 0:19)

surv.grade2.exp<-predict(whl.exp,
                          data.val = data.frame(grade=2),
                          time.pts = 0:19)

#survival probability estimates at 10 years
surv.grade1.exp$results[10,]
```

	time.pts	as.factor(grade)	hazard	hazard.inf	hazard.sup	surv
10	10	1	0.004425163	0.00359919	0.005440688	0.9567132
			surv.inf	surv.sup		
10			0.9470467	0.9646481		

```
surv.grade2.exp$results[10,]
```

```

time.pts as.factor(grade) hazard hazard.inf hazard.sup surv
10      10              2 0.008809281 0.006895095 0.01125487 0.9156759
      surv.inf surv.sup
10 0.8935538 0.9333725

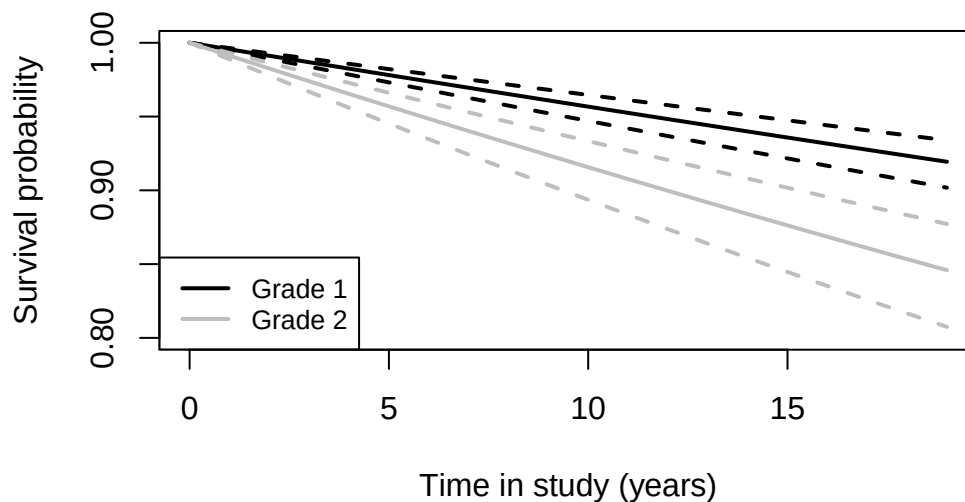
```

```

#Plot estimated survival curves
plot(0:19,c(1,surv.grade1.exp$results$surv),type="l",
     col="black",lwd=2,ylim=c(0.8,1),
     xlab="Time in study (years)",ylab="Survival probability")
points(0:19,c(1,surv.grade2.exp$results$surv),type="l",
       col="grey",lwd=2)
points(0:19,c(1,surv.grade1.exp$results$surv.inf),type="l",
       col="black",lwd=2,lty=2)#CI
points(0:19,c(1,surv.grade1.exp$results$surv.sup),type="l",
       col="black",lwd=2,lty=2)#CI
points(0:19,c(1,surv.grade2.exp$results$surv.inf),type="l",
       col="grey",lwd=2,lty=2)#CI
points(0:19,c(1,surv.grade2.exp$results$surv.sup),type="l",
       col="grey",lwd=2,lty=2)#CI

legend("bottomleft",
      c("Grade 1","Grade 2"),
      col=c("black","grey"),lwd=2,cex=0.8)

```



SOLUTION

- Under the exponential model the survival probability at time t is given by $S(t|X) = \exp\{-t\lambda e^{\beta X}\}$. The probability of survival beyond 10 years for a person in grade 1 is therefore estimated to be $\exp\{-10 \times 0.0044\} = 0.957$. The probability of survival beyond 10 years for a person in grade 1 is therefore estimated to be $\exp\{-10 \times 0.0044 \times 1.99\} = 0.916$.

Question 4

Change to using age as the time scale (accounting for delayed entry into the study) and refit the exponential model. Compare your results with those found when using time-in-study as the timescale, and explain the results.

SOLUTION

```
#Fitting exponential model using mexhaz, with age as time scale
whl.exp2<-mexhaz(Surv(time=as.numeric(timein)/365.25,
                      time2=as.numeric(timeout)/365.25,
                      event=chd,
                      origin=as.numeric(timebth))~as.factor(grade),
                 base="pw.cst",data=whl,print.level = 0)
```

Data

Name	N.Obs.Tot	N.Obs	N.Events	N.Clust
whl	1677	1677	154	1

Details

Iter	Eval	Base	Nb.Leg	Nb.Aghq	Optim	Method	Code	LogLik	Total.Time
8	12	pw.cst	20	10	nlm	---	1	-944.6851	0.01

```
summary(whl.exp2)
```

Call:

```
mexhaz(formula = Surv(time = as.numeric(timein)/365.25, time2 = as.numeric(timeout)/365.25,
                      event = chd, origin = as.numeric(timebth)) ~ as.factor(grade),
        data = whl, base = "pw.cst", print.level = 0)
```

Coefficients:

	Estimate	StdErr	z.value	p.value
[0,2.59e+04]	-5.42045	0.10541	-51.4229	< 2.2e-16 ***
as.factor(grade)2	0.68850	0.16351	4.2107	2.546e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Note: For a piecewise constant baseline hazard, the coefficients correspond to the logarithm of the hazard on each time interval.

Hazard ratios (for proportional effect variables):

	Coef	HR	CI.lower	CI.upper
as.factor(grade)2	0.6885	1.9907	1.4449	2.7428

log-likelihood: -944.6851 (for 2 degree(s) of freedom)

number of observations: 1677, number of events: 154

Using age as the time scale gives the same parameter estimates as when using time-in-study as the time scale here. This is because the hazard is constant over time in an exponential model, so the time-scale does not have any impact on the estimates. In an exponential model the estimated hazard in each group is $\sum_i \delta_i / \sum_i t_i$, where $\sum_i \delta_i$ is the number of events, and $\sum_i t_i$ is the total time at risk. Neither of these are altered by changing to the age time scale, as long as the left truncation is accounted for. Hence the parameter estimates are unaffected.

Question 5

Go back to the time-in-study timescale for this question and all subsequent questions. By fitting an exponential distribution we are assuming that the hazard rate does not change over time. Because this may not be a reasonable assumption we investigate fitting a Weibull model.

- Write down the form of the hazard model.
- Fit the Weibull model by using `base="weibull"` in `mexhaz`.
- What are the parameter estimates? Interpret the hazard ratio estimate.
- Use the results to compare the fit of the exponential and Weibull models.

SOLUTION

```
whl.wei<-mexhaz(Surv(time=time,event=chd)~as.factor(grade),
               base="weibull",data=whl,print.level = 0)
```

Data

Name	N.Obs.Tot	N.Obs	N.Events	N.Clust
------	-----------	-------	----------	---------

```
whl      1677  1677      154      1
```

Details

```
Iter Eval   Base Nb.Leg Nb.Aghq Optim Method Code   LogLik Total.Time
   8   11 weibull    20     10   nlm    ---     1 -937.7077      0.01
```

```
summary(whl.wei)
```

Call:

```
mexhaz(formula = Surv(time = time, event = chd) ~ as.factor(grade),
       data = whl, base = "weibull", print.level = 0)
```

Coefficients:

```
              Estimate      StdErr  z.value  p.value
logLambda      -6.446303   0.320810 -20.0938 < 2.2e-16 ***
logRho           0.305864   0.077708   3.9361 8.283e-05 ***
as.factor(grade)2 0.712250   0.163624   4.3530 1.343e-05 ***
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Hazard ratios (for proportional effect variables):

```
              Coef      HR CI.lower CI.upper
as.factor(grade)2 0.7122 2.0386   1.4793   2.8093
```

log-likelihood: -937.7077 (for 3 degree(s) of freedom)

number of observations: 1677, number of events: 154

```
confint(whl.wei)
```

```
              2.5 %      97.5 %
logLambda      -7.075079 -5.8175266
logRho           0.153559  0.4581688
as.factor(grade)2 0.391552  1.0329472
```

- The hazard model is $h(t|X) = \kappa \lambda t^{\kappa-1} e^{\beta X}$.
- The parameter estimates are: $\lambda = e^{-6.446} = 0.0016$, $\hat{\kappa} = e^{0.306} = 1.358$, $\hat{\beta} = 0.712$.
- The hazard ratio estimate is $e^{\hat{\beta}} = 2.04$, which has 95% CI (1.48, 2.81). Those in job grade 2 are estimated to have a hazard that is just over two times the hazard of those in job grade 1. The 95% confidence interval excludes 1. There is strong evidence of an association between job grade and the hazard.

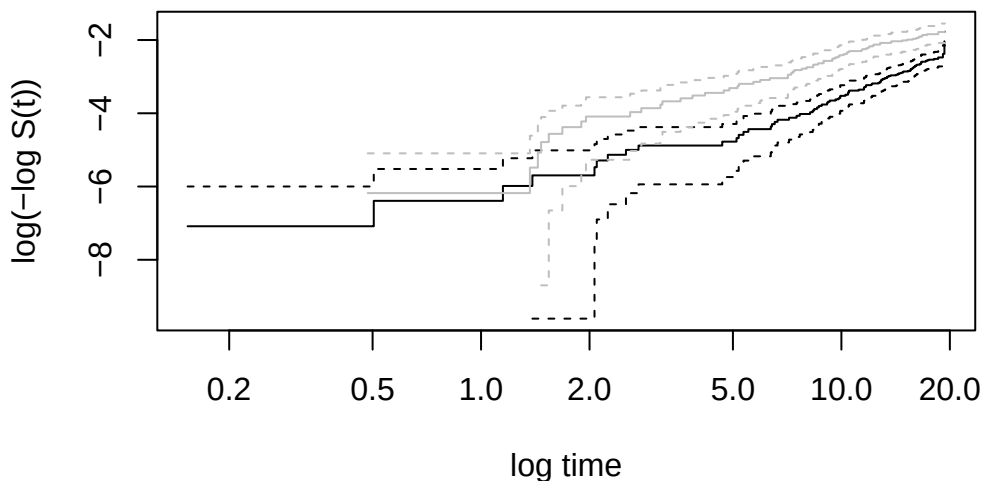
- The estimate of the shape parameter κ is 1.358. The Weibull distribution with $\kappa = 1$ is equivalent to the exponential distribution. We can therefore test the null hypothesis that $\kappa = 1$ (or equivalently $\log \kappa = 0$) in order to test the null hypothesis that the hazard rate does not change over time. The estimate of $\log \kappa$ shown in the output has very small p-value and 95% CI (0.391, 0.458) that excludes the null value 0. There is strong evidence against the null hypothesis that the hazard rate does not change over time, indicating that the Weibull model provides a better fit to the data compared to the exponential model.

Question 6

Use the code below to generate a plot that helps to investigate whether you expect the Weibull model fitted above to be appropriate.

- What is being plotted?
- Does the Weibull model appear to provide a good fit?

```
whl.km <- survfit(Surv(time=time,event=chd)~as.factor(grade),
                  data=whl)
plot(whl.km,conf.int=T,fun="cloglog",
     col=c("black","grey"),
     xlab="log time", ylab="log(-log S(t))")
```



SOLUTION

- If the Weibull model is appropriate, a plot of $\log H(t|x) = \log\{-\log S(t|X)\}$ against $\log t$ is linear in each job grade group, and the two lines are parallel. In the plot we can see that (after the initial period where the confidence intervals are wide) the lines are both straight, and are approximately parallel. This suggests the Weibull model is reasonable.

Question 7

The age at which individuals entered the study may have an important part to play in explaining the time to event.

- Add age as an additional variable in the Weibull model.
- Interpret the hazard ratios for job grade and for age.
- What effect does adjusting for age at entry to the study have on the hazard ratio for job grade? Can you explain why this might happen?
- Based on this model, obtain estimated survivor curves for individuals in job grades 1 and 2, and aged 50 and 60. You can use the code below to obtain the estimates. Interpret the plots.

```
#Weibull model including grade and age
whl.wei.age <- mexhaz(Surv(time=time,event=chd)~as.factor(grade)+agein,
                     base="weibull",data=whl,print.level = 0)

#estimated survivor curves for individuals with
# job grade 1 and 2, aged 50 and 60.
#Shown for 1 individual - you can expand this for the others.
newdata<-expand.grid(grade=c(1,2),agein=c(50,60))
pred.wei.1<-predict(whl.wei.age,data.val=newdata[1,], time.pts=seq(0,19,1))
```

SOLUTION

```
#Weibull model including grade and age
whl.wei.age <- mexhaz(Surv(time=time,event=chd)~as.factor(grade)+agein,
                     base="weibull",data=whl,print.level = 0)
```

Data

Name	N.Obs.Tot	N.Obs	N.Events	N.Clust
whl	1677	1677	154	1

Details

Iter	Eval	Base	Nb.Leg	Nb.Aghq	Optim	Method	Code	LogLik	Total.Time
8	11	weibull	20	10	nlm	---	1	-905.4786	0.03

```
summary(whl.wei.age)
```

Call:

```
mexhaz(formula = Surv(time = time, event = chd) ~ as.factor(grade) +  
      agein, data = whl, base = "weibull", print.level = 0)
```

Coefficients:

	Estimate	StdErr	z.value	p.value
logLambda	-12.056256	0.817230	-14.7526	< 2.2e-16 ***
logRho	0.345756	0.076610	4.5132	6.387e-06 ***
as.factor(grade)2	0.237951	0.174758	1.3616	0.1733
agein	0.104326	0.013416	7.7762	7.550e-15 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Hazard ratios (for proportional effect variables):

	Coef	HR	CI.lower	CI.upper
as.factor(grade)2	0.2380	1.2686	0.9007	1.7869
agein	0.1043	1.1100	1.0812	1.1395

log-likelihood: -905.4786 (for 4 degree(s) of freedom)

number of observations: 1677, number of events: 154

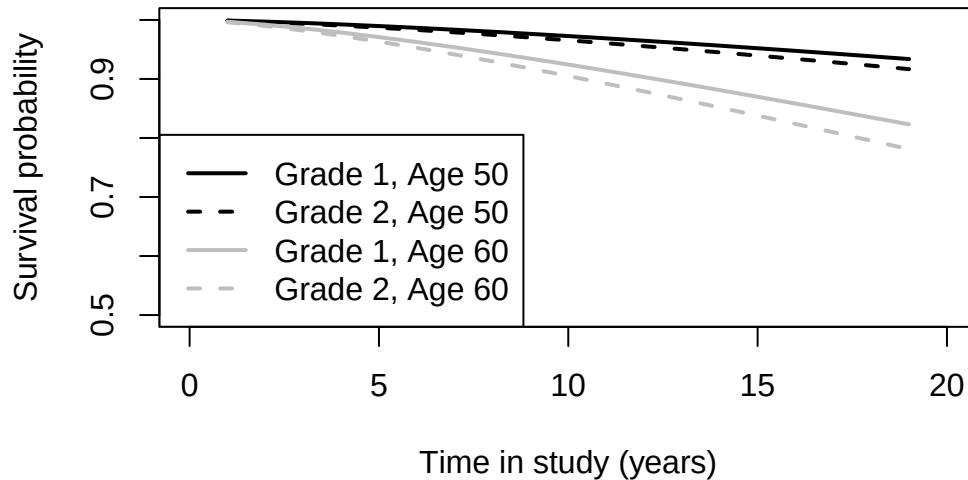
```
#estimated survivor curves for individuals with  
# job grade 1 and 2, aged 50 and 60  
  
newdata<-expand.grid(grade=c(1,2),agein=c(50,60))  
  
pred.wei.1<-predict(whl.wei.age,data.val=newdata[1,], time.pts=seq(0,19,1))  
pred.wei.2<-predict(whl.wei.age,data.val=newdata[2,], time.pts=seq(0,19,1))  
pred.wei.3<-predict(whl.wei.age,data.val=newdata[3,], time.pts=seq(0,19,1))  
pred.wei.4<-predict(whl.wei.age,data.val=newdata[4,], time.pts=seq(0,19,1))  
  
plot(pred.wei.1$results$time,pred.wei.1$results$surv,type="l",  
      lwd=2,lty=1,col="black",  
      xlab="Time in study (years)",ylab="Survival probability",  
      xlim=c(0,20),ylim=c(0.5,1))  
lines(pred.wei.2$results$time,pred.wei.2$results$surv,type="l",  
      lwd=2,lty=2,col="black")  
  
lines(pred.wei.3$results$time,pred.wei.3$results$surv,type="l",  
      lwd=2,lty=1,col="grey")  
lines(pred.wei.4$results$time,pred.wei.4$results$surv,type="l",
```

```

lwd=2,lty=2,col="grey")

legend("bottomleft",c("Grade 1, Age 50","Grade 2, Age 50",
                      "Grade 1, Age 60","Grade 2, Age 60"),
      lwd=2,lty=c(1,2,1,2),
      col=c("black","black","grey","grey"))

```



- The hazard ratio for age is 1.11 (95% CI 1.08 to 1.14), conditional on grade. The interpretation is that the hazard for CHD death increases by 11% for each additional year of age at entry to the study, conditional on job grade. Age is strongly associated with the hazard, conditional on job grade.
- The estimated hazard ratio for grade is 1.27 (95% CI 0.90 to 1.79), after conditioning on age at entry to the study. The point estimate means that being in grade 2 is associated with a 27% increase in the hazard compared to being in grade 1, conditional on age. However, the confidence interval includes the null value of 1.
- Before adjustment for age, the hazard ratio for job grade was 2.04. After conditioning on age the hazard ratio is much closer to 1 and there is no evidence of an association between job grade and the hazard. One reason for the change in the hazard ratio is that age is associated with both job grade and the hazard, i.e. it is a confounder.
- Looking at the estimated survival curves, the curve for an individual in job grade 1 who entered the study at age 50 indicates the best survival, and an individual in job grade 2 and aged 60 has the worst survival (among our example individuals, that is). Given age, survival is seen to be worse for those in job grade 2 compared to job grade 1. And given job grade, survival is seen to be worse with older age at entry to the study.

Question 8

The Weibull model allows the baseline hazard to change over time. A different approach is to split the follow up time into periods and fit an exponential model within each period, but allowing hazard ratios to be the same in each period. In other words, to allow the baseline hazard to differ in the different periods. This called a *piecewise exponential model*. It can then be investigated whether the baseline hazard changes across the periods. One way to do this is to create a record for each individual within each time period up to their event or censoring time. This approach of fitting exponential models within time bands is sometimes called *Lexis expansion*.

- Use the code below to set the data up for fitting a piecewise exponential model, and then to fit the model.
- Interpret the results. What is the baseline hazard in the four time periods? Notice that age has been centered at age 60 to help make the interpretation more sensible.
- Compare the results from this model with those from the Weibull model fitted earlier.

```
#Take a look at two example individuals
whl[whl$id %in% c(5001,5350),c("id","timein","timeout","time","chd")]

#Split the data into several rows per person
whl.split<- survSplit(Surv(time,chd)~., whl, cut=c(5,10,15,20),
                      start="tstart",end="time", episode="period")

#Take a look at what has happened to the data for the two example individuals
whl.split[whl.split$id %in% c(5001,5350),c("id","tstart","time","chd","period")]

#fit a piecewise exponential model, with age centered at age 60
whl.pw<-mexhaz(Surv(time=time,event=chd)~
               as.factor(grade)+I(agein-60)+as.factor(period),
               base="pw.cst",data=whl.split,print.level = 0)
summary(whl.pw)
```

SOLUTION

```
#Take a look at two example individuals
whl[whl$id %in% c(5001,5350),c("id","timein","timeout","time","chd")]
```

	id	timein	timeout	time	chd
1	5001	1967-09-13	1987-01-30	19.38125	0
44	5350	1967-09-20	1980-07-12	12.81040	1

```
#Split the data into several rows per person
whl.split<- survSplit(Surv(time,chd)~., whl, cut=c(5,10,15,20),
                      start="tstart",end="time", episode="period")

#Take a look at what has happened to the data for the two example individuals
whl.split[whl.split$id %in% c(5001,5350),
          c("id","tstart","time","chd","period")]
```

	id	tstart	time	chd	period
1	5001	0	5.00000	0	1
2	5001	5	10.00000	0	2
3	5001	10	15.00000	0	3
4	5001	15	19.38125	0	4
168	5350	0	5.00000	0	1
169	5350	5	10.00000	0	2
170	5350	10	12.81040	1	3

```
#fit a piecewise exponential model, with age centered at age 60
whl.pw<-mexhaz(Surv(tstart,time,event=chd)~
               as.factor(grade)+I(agein-60)+as.factor(period),
               base="pw.cst",data=whl.split,print.level = 0)
```

Data

	Name	N.Obs.Tot	N.Obs	N.Events	N.Clust
whl.split		6167	6167	154	1

Details

Iter	Eval	Base	Nb.Leg	Nb.Aghq	Optim	Method	Code	LogLik	Total.Time
8	11	pw.cst	20	10	nlm	---	1	-904.5096	0.2

```
summary(whl.pw)
```

Call:

```
mexhaz(formula = Surv(tstart, time, event = chd) ~ as.factor(grade) +
        I(agein - 60) + as.factor(period), data = whl.split, base = "pw.cst",
        print.level = 0)
```

Coefficients:

Estimate	StdErr	z.value	p.value
----------	--------	---------	---------

```

[0,19.4]          -5.236273  0.217443 -24.0811 < 2.2e-16 ***
as.factor(grade)2  0.238666  0.174740  1.3658  0.171990
I(agein - 60)      0.104786  0.013438  7.7977 6.217e-15 ***
as.factor(period)2 0.629910  0.242499  2.5976  0.009388 **
as.factor(period)3 0.781003  0.242875  3.2157  0.001301 **
as.factor(period)4 1.072444  0.257181  4.1700 3.046e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Note: For a piecewise constant baseline hazard, the coefficients correspond to the logarithm of the hazard on each time interval.

Hazard ratios (for proportional effect variables):

	Coef	HR	CI.lower	CI.upper
as.factor(grade)2	0.2387	1.2696	0.9014	1.7881
I(agein - 60)	0.1048	1.1105	1.0816	1.1401
as.factor(period)2	0.6299	1.8774	1.1672	3.0198
as.factor(period)3	0.7810	2.1837	1.3566	3.5150
as.factor(period)4	1.0724	2.9225	1.7654	4.8380

log-likelihood: -904.5096 (for 6 degree(s) of freedom)

number of observations: 6167, number of events: 154

- The piecewise exponential hazard model being fitted can be written $h(t|X) = \lambda_1 \exp\{\gamma_2 I(5 < t \leq 10) + \log \gamma_3 I(10 < t \leq 15) + \gamma_4 I(15 < t \leq 20) + \beta_1 \text{grade} + \beta_2 (\text{age} - 60)\}$, where *grade* takes value 0 for those in job grade 1 and value 1 for those in grade 2.
- The baseline hazard refers to the hazard for a person in job grade 1, aged 60. The estimated baseline hazard in time period 1 ($0 < t \leq 5$) is $\hat{\lambda}_1 = \exp(-5.236) = 0.0053$. The estimated baseline hazards in periods 2 ($5 < t \leq 10$), 3 ($10 < t \leq 15$) and 4 ($15 < t \leq 20$) are respectively: $\hat{\lambda}_1 e^{\hat{\gamma}_2} = 0.0053 \times 1.8774 = 0.0100$, $\hat{\lambda}_1 e^{\hat{\gamma}_3} = 0.0053 \times 2.1837 = 0.0116$, $\hat{\lambda}_1 e^{\hat{\gamma}_4} = 0.0053 \times 2.9225 = 0.0155$.
- The hazard ratios for periods 2, 3 and 4 relative to period 1 indicate that the baseline hazard gets higher over time in the study.
- The hazard ratios for grade 2 versus grade 1 is 1.27 (95% CI 0.90 to 1.79) and the HR for age is 1.11 (95% CI 1.08 to 1.14). These estimates are similar to those found using the Weibull model.